

M-Stream 2.2

* MStream_ChangeLog

M-Stream data has packet structure. Large packets may be split to fragments with same Packet ID to fit in encapsulating frame.

M-Stream 2.0 defines M-Stream Header, Event Header and User Payload data objects. In fragmentation process, M-Stream Header is persistent and should be present in all fragments. Event Header should be present in first fragment only.

M-Stream Header			
word #	byte offset	bits	description
0	0	31:24	Device ID
		23:18	Flags
		17:16	Subtype
		15:0	Fragment length
1	4	31:16	Packet ID
		15:0	Fragment offset (bytes)

Flags bits	
bit	value
5	LF(last fragment)
4	EVC (Event Complete)
3	FIN (<i>not used yet</i>)
2	SYN (<i>not used yet</i>)
1	RST (<i>not used yet</i>)
0	ACK

Fragment length is defined as number of bytes starting at offset 8. For M-Stream frame of size N 32-bit words, fragment length value is $4 \times (N-2)$.

Fragmentation supports maximum of $2^{16} = 64$ kByte data payloads.

Packet ID is 16-bit number (0-65535).

Flags:

- LF - last part of current packet ($\langle \text{Fragment length} \rangle + \langle \text{Fragment offset} \rangle = \langle \text{Full length of this packet} \rangle$)
- EVC - last part of current event (One sets with LF flag for last channel in current event)
- ACK - acknowledge flag: is set in MStream receiver program to inform device of receiving package which contains fragment with corresponding FragmentId and FragmentOffset

Subtype 0: Trigger and User Data

M-Stream Header			
word #	byte offset	bits	description
0	0	31:24	Device ID
		23:18	Flags
		17:16	Subtype (0)
		15:0	Fragment length, 4*(N-2)
1	4	31:16	Packet ID
		15:0	Fragment offset (bytes)
Subtype 0 Header			
2	8	31:0	Device serial number
3	12	31:24	Subtype and custom bits
		23:0	Event number
M-Stream Subtype 0 Payload			
4	16	31:0	Event timestamp, TAI seconds
5	20	31:2	Event timestamp, TAI nanoseconds
		1:0	Event timestamp, TAI flags
6	24	31:0	User defined payload
...			
N-1	4*(N-1)	31:0	

TAI flags: 2 - timecode is valid, otherwise invalid.

Subtype 1: User Data, Channel-Based Readout

M-Stream Header			
word #	byte offset	bits	description
0	0	31:24	Device ID
		23:18	Flags
		17:16	Subtype (1)
		15:0	Fragment length, 4*(N-2)
1	4	31:16	Packet ID

		15:0	Fragment offset (bytes)
Subtype 1 Header			
2	8	31:0	Device serial number
3	12	31:24	Channel number
		23:0	Event number
M-Stream Subtype 1 Payload			
4	16	31:0	User defined payload
...			
N-1	4*(N-1)	31:0	

Subtype 2: User Data, Time Slice Based Readout

M-Stream Header			
word #	byte offset	bits	description
0	0	31:24	Device ID
		23:18	Flags
		17:16	Subtype (1)
		15:0	Fragment length, 4*(N-2)
1	4	31:16	Packet ID
		15:0	Fragment offset (bytes)
Subtype 2 Header			
2	8	31:0	Device serial number
3	12	31:0	reserved
M-Stream Subtype 2 Payload			
4	24	31:0	User defined payload
...			
N-1	4*(N-1)	31:0	

Acknowledge

Acknowledge package is similar to standard M-Stream package. First 32-bit word should have acknowledge bit up to 22nd bit, as described in *Flag* field in M-Stream 2.0. All following words will be proceeded as *Packet Id* and *Fragment Offset* pairs. One package can contain one acknowledge pair (by default) or several (request M-Stream configuration register in M-Link CSR).

MLink MStream RawData